Micro Set 4. Positive Externalities & Merit Goods — Model Answer Scheme

Question 1 [2 marks]

A **merit good** is a good or service that generates positive externalities in consumption and/or is subject to information failure, leading to under-consumption in a free market. *[1 mark.]*

Evidence from Extract 1.1: (i) university education generates positive externalities through "skills accumulation, research output, and social cohesion," and (ii) consumers exhibit "a time-inconsistency problem where individuals systematically defer self-investment that would benefit them in the long run" — i.e. information failure regarding long-term private returns. *[1 mark for two distinct features.]*

> **Examiner Note.** Merit-good characterisation requires both pillars: positive externality *and* information failure. Naming only one earns 1 mark.

Question 2 [2 marks]

$$\%\Delta = \frac{6.4 - 5.2}{5.2} \times 100\% = +23.1\%$$

[1 mark working, 1 mark answer.] Polyclinic subsidised visits rose by approximately 23 per cent between 2019 and 2024, consistent with the Ministry's statement in Extract 1.1.

Question 3 [4 marks]

Why university education would be under-consumed in a free market.

Extract 1.1 identifies two distinct sources of under-consumption.

First, positive externality in consumption. Education generates social benefits beyond the private student: "skills accumulation, research output, and social cohesion," contributing to higher aggregate productivity and tax revenues. The marginal social benefit (MSB) exceeds the marginal private benefit (MPB) by the marginal external benefit (MEB). Free-market consumption equates MPB with MPC at Q_m, but the socially optimal level is Q^* where MSB = MSC. Hence $Q_m < Q^*$. *[2 marks.]*

Second, information failure. Students at the consumption decision point are typically 18–19 years old and may systematically under-estimate the long-term private returns to higher education (lifetime earning premium, career options, networks). This further reduces MPB-perceived below true MPB, exacerbating under-consumption. *[1 mark.]*

The combination of MEB wedge plus information failure makes university education a classic merit good with under-allocation $|Q_m - Q^*|$ that the free market cannot self-correct without intervention. *[1 mark.]*

> **Examiner Note.** Top answers cleanly distinguish the two sources of failure. Conflating them or mentioning only one caps at 2–3 marks.

Question 4 [4 marks]

Why information failure weakens vaccination subsidies.

A vaccination subsidy reduces the financial cost of getting vaccinated. If the only barrier were financial, the subsidy would directly shift consumption from Q_m toward Q^*. *[1 mark.]*

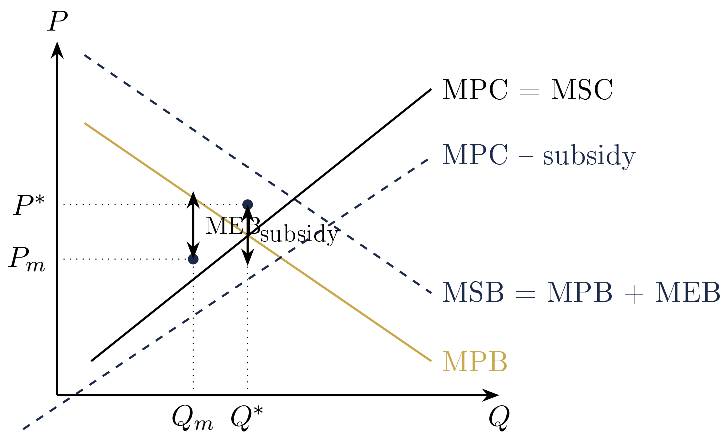
However, Extract 1.1 reports that "many adults below 65 do not perceive themselves as at risk and discount the social benefit of reducing transmission." This is information failure of two kinds: (a) personal risk underestimation (consumers misperceive their own marginal private benefit), and (b) external-benefit ignorance (consumers do not consider their role in herd immunity). *[2 marks for dual-failure characterisation.]*

The subsidy lowers price but does not correct the misperception. Even at zero price, if consumers do not value the vaccine, demand remains below the socially optimal level. Empirically, the vaccination rate rose to 42% in 2021 (Covid-driven) but fell back to 33–35% by 2024 (Table 1.1), demonstrating that price-based intervention alone is insufficient. *[1 mark for data anchor.]*

> **Examiner Note.** The key insight: subsidies correct *price* but not *information*. Without recognising this distinction, the answer becomes generic.

Question 5 [6 marks]

Subsidy to correct under-consumption of higher education.



In the diagram, MPB is private demand. MSB = MPB + MEB, with the MEB wedge reflecting the positive externalities (productivity, innovation, social cohesion). The free-market equilibrium occurs at Q_m, where MPB = MPC. The socially optimal output is Q^*, where MSB = MSC. *[2 marks for diagram labelled with MSB wedge.]*

A subsidy paid to universities (or directly to students) of magnitude equal to MEB at Q^* shifts the effective MPC curve downward (MPC − subsidy). The new private equilibrium occurs at Q^*, where MPB now intersects the subsidised cost curve. *[2 marks for subsidy-shift mechanism.]*

The price paid by consumers falls from P_m to a lower student-paid price, while suppliers receive P_m+ subsidy. The quantity consumed rises from Q_m to Q^*, eliminating the under-consumption and restoring allocative efficiency. *[1 mark for price-and-quantity outcomes.]*

Table 1.1 confirms the empirical pattern: as mean subsidy per student rose from S$18000 (2019) to S$25500 (2024), the cohort participation rate rose from 36% to 42%, consistent with the subsidy successfully expanding consumption. *[1 mark for data anchor.]*

> **Examiner Note.** The diagram must label MSB *and* the subsidy-shifted MPC. Without both, the mechanism is incomplete.

Question 6 [6 marks]

Why HPB combines subsidies with information campaigns.

Reason 1: Subsidies alone leave information failure uncorrected. As established in Q4, vaccines are free or heavily subsidised, yet vaccination uptake remains around 33–35% (Extract 1.1). The barrier is not financial but perceptual: consumers under-estimate personal risk and the social benefit. Information campaigns directly address this by correcting misperceptions about influenza risk, vaccine efficacy, and the externality of herd immunity. *[2 marks.]*

Reason 2: Behavioural and search costs. Even when consumers understand the vaccine's value, getting vaccinated involves a time cost (researching where to go, scheduling, travelling). Workplace-partnership programmes (Extract 1.1) reduce this search cost by bringing vaccination into the workplace, leveraging the social-norm effects of group participation. *[2 marks.]*

Reason 3: Synergistic effect. Information campaigns raise demand by shifting MPB toward true MPB. Subsidies expand the affordable supply at lower cost. The two policies are complements: an informed consumer who also faces low cost is far more likely to vaccinate than one facing only one of the two interventions. *[2 marks.]*

> **Examiner Note.** The strongest answers explicitly frame the policies as *complements* (multiplicative) rather than substitutes (additive). This is the merit-good policy mix principle.

End of Set 4 — Model Answers